MATT Improvement Senior Design Project

Meeting Minutes — October 22, 2026

Meeting Duration: 48 minutes, 38 seconds

Meeting Purpose: Meet with an experienced MAT end user to walk through the application from an assessor's perspective, identify usability problems encountered during live exercises, gather feature requests, and provide the team with access to a MAT test instance.

Attendees

- Samuel Bricking — Project advisor
- Nathan “Mac” McAvoy — MAT end user and exercise assessor (guest)
- Cooper Orlowski
- Sam Anderson
- Luke Link
- Srithan Mopuru
- Noah Nguyen
- Xuhui Zou

---

1. Meeting Background

This meeting followed the October 15 session with Doug Disbennett and Gloria Yeomans, which covered MAT's administrative and back-end requirements. As recommended in that meeting, this session focused on feedback from an actual user of the application during live exercises.

Mac has worked with MAT for a little over six years, originally through the military's Cyber Shield exercise and later supporting assessments for the Cyber Guardian exercise. He described substantial frustration with the application, both his own and that of other assessors.

Samuel shared his screen and navigated a MAT event created specifically for the capstone team, so that previous exercise data would not be exposed. Mac directed the walkthrough and identified the problems he considered most important.

2. How MAT Is Used During Exercises

Assessor Experience

- Most assessors are volunteers who arrive with very little experience using MAT.
- Many assessors are not familiar with cyber exercises or do not have a strong cybersecurity or IT background.
- Preparation typically consists of four one-hour meetings spaced about a month apart, followed by roughly three-quarters of a day of training the day before the event.
- Assessors usually become comfortable with the tool around the time the exercise ends, and returning assessors often forget how to use it after a year away.

Play Cards

Play cards drive the exercise. Each play card describes what is supposed to happen, who is involved, and which measurement codes are evaluated for that attack or event. During the exercise, the White Cell announces which play card is starting, and assessors evaluate the associated measures.

- All teams are typically assessed on the same tasks, but teams do not complete play cards in the same order, so assessors for different teams may be working on different play cards at the same time.
- Measurement codes on a play card are not listed in order, and the same measurement code can appear on multiple play cards. Teams may therefore have several opportunities to complete a task.
- The first number in a measurement code identifies the core task and the second identifies the day; deeper levels of the hierarchy become difficult to read and remember.
- Real play cards contain information that cannot be shared with the team, so a blank or redacted example is needed.

Measurement Terminology

METLs represent the overall tasks and are made up of Measures of Performance (MOPs) and Measures of Effectiveness (MOEs). MOPs are essentially an evaluation of whether the team performed the action. Samuel will provide reference documentation explaining the acronyms.

3. Assessor Pain Points and Requested Improvements

Team-Based Permissions

- During an exercise, every assessor can see the assessments for all teams. Assessors have filled out evaluations for the wrong team as a result, especially on the first day.
- Assessors can also view other teams' assessment wheels, and some have been persuaded to show teams how other teams are performing.
- MAT does support permissions, but the settings are buried in the back end and have behaved inconsistently: users either see nothing or still see everything after restrictions are applied. As a result, they are often not used.

Requested improvement: Restrict assessors to their assigned team, ideally through a simple setting on the user account, while allowing assessment leads to view all teams. Mac noted that the best way to eliminate user error is to remove the choice.

Assessment Wheel

Mac confirmed that the wheel is popular and must be preserved, but identified several usability problems:

- Segment labels are small and difficult to read.
- There is no obvious way to move back a level or reset the wheel after drilling down, which makes it awkward to use while presenting results to a team.
- At the deepest drill-down level, the wheel displays no useful information and offers no actions.
- To evaluate an incomplete measure, assessors must drill into the wheel, write down the measurement code (often on sticky notes), back out, open Measure Evaluations, search for the code, and then edit the evaluation.

Requested improvements:

1. Add a legend beside the wheel or hover pop-ups to identify segments.
2. Add back and forward navigation buttons.
3. Allow users to click a measure in the wheel to view its status and the assessor's comments, or open the evaluation directly. Samuel considered this a likely low-effort, high-impact feature and a good place to start.

Measure Evaluations View

- The view presents one large list of every measure, so assessors must scan for the measures on the current play card. This extra step frequently leads to evaluations being entered in the wrong place and then copied and deleted.
- The view repeats information such as inject titles and IDs, but the status column only shows values such as started or complete rather than what the assessor actually wrote.

Requested improvement: Allow the list to be grouped or sorted by play card, or default to sorting by day, and display the assessor's written input in the list.

Play Card View

- Create a view within the application that mirrors the layout of a play card.
- Allow users to click a measurement code on the play card to open its evaluation directly, eliminating the need to switch between screens or monitors.
- Allow play cards to be viewed, edited, and exported within the application so the process lives in one place.
- This would also support mid-exercise changes. Currently, staff repurpose unused existing measures when they need to change a play card or evaluate something unplanned.

Assessor Guidance for Each Measure

Assessors often work with the evaluation screen, the play card, a supplemental document, and Teams messages open simultaneously. The supplemental document, prepared by the scenario team, explains what assessors should look for for each measure. For example, the participants who should be alerted differ between a website breach and an account breach. MAT currently has no place to display this information.

Requested improvement: Add a short guidance field directly beneath each measure's description. The field should be editable by an administrator or elevated account, pushed to all teams, and updatable during the exercise, even immediately before a play card begins.

Team-Specific Task Lists

METLs are currently fixed for an event. In one year, a team moved from certification to training but remained assigned to the certification measures. This reinforces the requirement from the October 15 meeting to assign task lists to individual teams instead of locking them to the event.

4. Reporting

- Teams receive a final report several weeks after the exercise. The data is exported as a CSV file and then manually converted into a readable PDF using other software.
- The conversion requires removing items that were not evaluated and other unnecessary information.
- The export viewed during the meeting did not include assessor comments.

Requested improvement: Provide a readable report export directly from the application that omits unevaluated items and includes all comments. Samuel expects the report format to require several iterations based on user feedback.

Mac summarized that ease of entering information and ease of reporting are the two most important capabilities the software must provide, because the purpose of MAT is to tell teams what they did well and what they need to improve.

5. Home Page and Navigation

In response to a question from Srithan about specific user interface changes, Mac described the home page experience:

- Assessors first land on a home page containing features they do not need, such as the vulnerability dashboard and findings.
- Assessors must use the Views drop-down menu to find the screens they actually use.
- The assessment wheel and Measure Evaluations are the two screens users constantly switch between.

Requested improvement: Place two large, obvious buttons on the home page for the assessment wheel and Measure Evaluations. Mac also noted that the user interface in general needs a visual refresh.

6. Application Access and Resources

Samuel received approval to give the team access to a MAT test instance shortly before the meeting. This resolves the access item from the October 15 meeting.

- A separate Capstone Course event has been created for the team.
- The team will share a single test account.
- The team is free to explore the application and is encouraged to find what does not work well.
- MAT is not consistently reliable. If the instance becomes unavailable, the team should notify Samuel.

Samuel will send the following resources after the meeting:

- Login information for the test instance
- The MAT user guide
- A quick reference guide explaining the acronyms and navigation
- Sample data for practicing the import process
- A blank or redacted play card

7. Decisions and Agreements

1. **Preserve the assessment wheel.** Users value the wheel, so it should be improved rather than replaced.
2. **Implement team-based permissions.** Assessors should only be able to view and evaluate their assigned team.
3. **Connect the wheel to evaluations.** Clicking from the wheel directly into a measure evaluation is a strong candidate for an initial feature.

4. **Organize evaluations around play cards.** The application should help assessors find the measures for the current play card quickly.
5. **Prioritize input and reporting.** Ease of entering assessments and producing readable reports are the highest-priority goals.
6. **Explore the test instance.** Team members will use the application directly before forming further questions.
7. **Plan for user testing.** Samuel anticipates user focus groups later in the project.

8. Action Items

Action Item	Responsible Party	Status
Send test instance login information, user guide, quick reference guide, and sample data	Samuel Bricking	Pending
Locate or redact a blank play card to share with the team	Samuel Bricking	Pending
Read the user guide and quick reference guide	Senior design team	Pending
Explore the MAT test instance and document usability problems	Senior design team	Pending
Notify Samuel if the MAT test instance becomes unavailable	Senior design team	Ongoing
Document the play card workflow and its relationship to measurement codes	Senior design team	Pending
Evaluate how team-based permissions could be implemented	Senior design team	Pending
Evaluate navigation from the assessment wheel to measure evaluations	Senior design team	Pending
Review the current report export and identify required changes	Senior design team	Pending
Arrange user focus groups later in the project	Samuel Bricking and senior design team	Future
Upload these meeting minutes to the team's GitHub repository	Cooper Orlowski	Planned

9. Open Questions for Follow-Up

The following questions remain unresolved and should be addressed as the project progresses:

- How are play cards structured, and how should they be stored and linked to measurement codes in the new application?
- What roles are needed for team-based permissions, such as assessor, assessment lead, and administrator?
- How should the wheel's labels and drill-down navigation be redesigned without losing its familiarity?
- What information should appear in a play card view, and who should be able to edit play cards?
- Who will write the per-measure assessor guidance, and when is it typically finalized before an exercise?
- What should the final report contain, and what format do teams and stakeholders prefer?
- How can comments be included in data exports?
- Which home page features are needed by non-assessor roles?
- How will feedback from newer assessors be gathered to complement experienced users' perspectives?

10. Next Steps

The immediate priority is for each team member to review the documentation Samuel provides and explore the MAT test instance. The team will meet again on Monday, and meetings with Samuel will continue regularly. Mac is available for follow-up questions.

Once the team is familiar with the application, it should document the play card and evaluation workflows, compare Mac's feedback with the requirements gathered on October 15, and prioritize improvements based on user impact and technical feasibility.

Key Takeaway: The team now has direct access to MAT and a clear picture of the problems assessors face during live exercises. The most significant issues are the lack of team-based restrictions, the disconnect between the assessment wheel and measure evaluations, the absence of play card organization, and the manual reporting process. Improving ease of input and ease of reporting should guide the project's design.